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Big Data GROUP WORK

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Descriptive Analysis of Cervical Cancer Risk Factors Using Big Data Techniques

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1. Abstract

This report presents a descriptive analysis of the Cervical Cancer (Risk Factors) dataset using Python. The primary aim is to explore relationships among various health indicators and their influence on cervical cancer diagnosis. The report covers data preprocessing, outlier detection, summary statistics, feature relationships, and visualizations. The final goal is to prepare a well-understood dataset and generate insights that can later be used for predictive modeling.

2. Introduction

Cervical cancer is one of the leading causes of cancer-related deaths among women, especially in low-resource settings. Early detection through risk factor analysis can significantly reduce mortality rates. This study uses a dataset from the UCI Machine Learning Repository to explore health-related factors that may contribute to cervical cancer risk.

3. Methodology

Dataset Description

The dataset consists of 858 samples and 36 features, including demographic, behavioral, and medical history data. The target variable is `Biopsy`, indicating whether the patient has been diagnosed with cervical cancer.

Tools and Libraries

- Python 3
- Pandas and NumPy for data handling
- Matplotlib and Seaborn for visualization
- More visualization using Power BI

Data Cleaning

Missing values were imputed using appropriate strategies. Binary categorical variables were encoded as 0 and 1. The dataset was then evaluated for consistency and completeness.

Outlier Detection and Removal

Outliers were detected using the IQR (Interquartile Range) method. The following features had significant outliers:

- Age: 8 outliers
- Number of sexual partners: 68 outliers
- First sexual intercourse: 41 outliers
- Num of pregnancies: 10 outliers
- Smokes (years): 136 outliers
- Smokes (packs/year): 136 outliers
- Hormonal Contraceptives (years): 114 outliers

- IUD (years): 200 outliers
- STDs (number): 184 outliers
- STDs: Number of diagnosis: 71 outliers
- STDs: Time since first diagnosis: 71 outliers
- STDs: Time since last diagnosis: 71 outliers

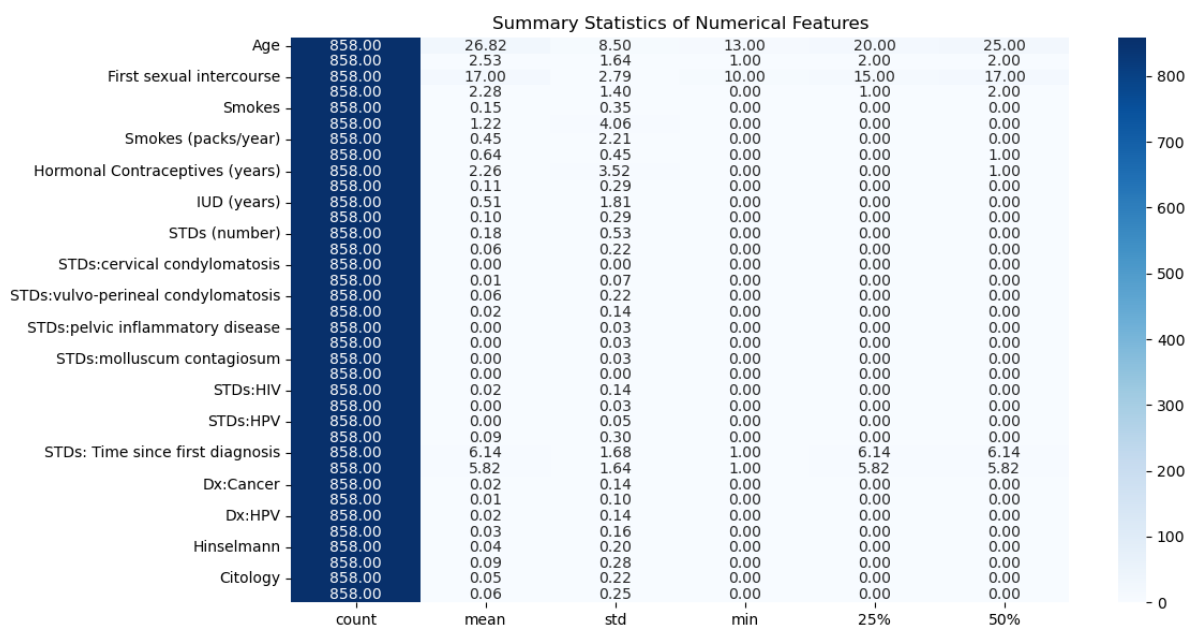
Due to the large number of outliers, values were removed based on percentile thresholds to preserve valuable data. After this filtering, the dataset was reduced to **385 rows**, improving reliability while maintaining critical cases.

4. Descriptive Analysis

Summary Statistics

Key insights from the summary statistics:

- The average age of patients is ~26.8 years, with the youngest being 13 and the oldest 84.
- Most women reported 2–3 sexual partners and an average first intercourse age of 17.
- Smoking behavior is relatively rare but shows high variation among smokers.
- Use of hormonal contraceptives is common (mean = 0.64), often for ~2.3 years.
- Diagnosed cancer-related conditions (CIN, HPV, etc.) have very low incidence.
-

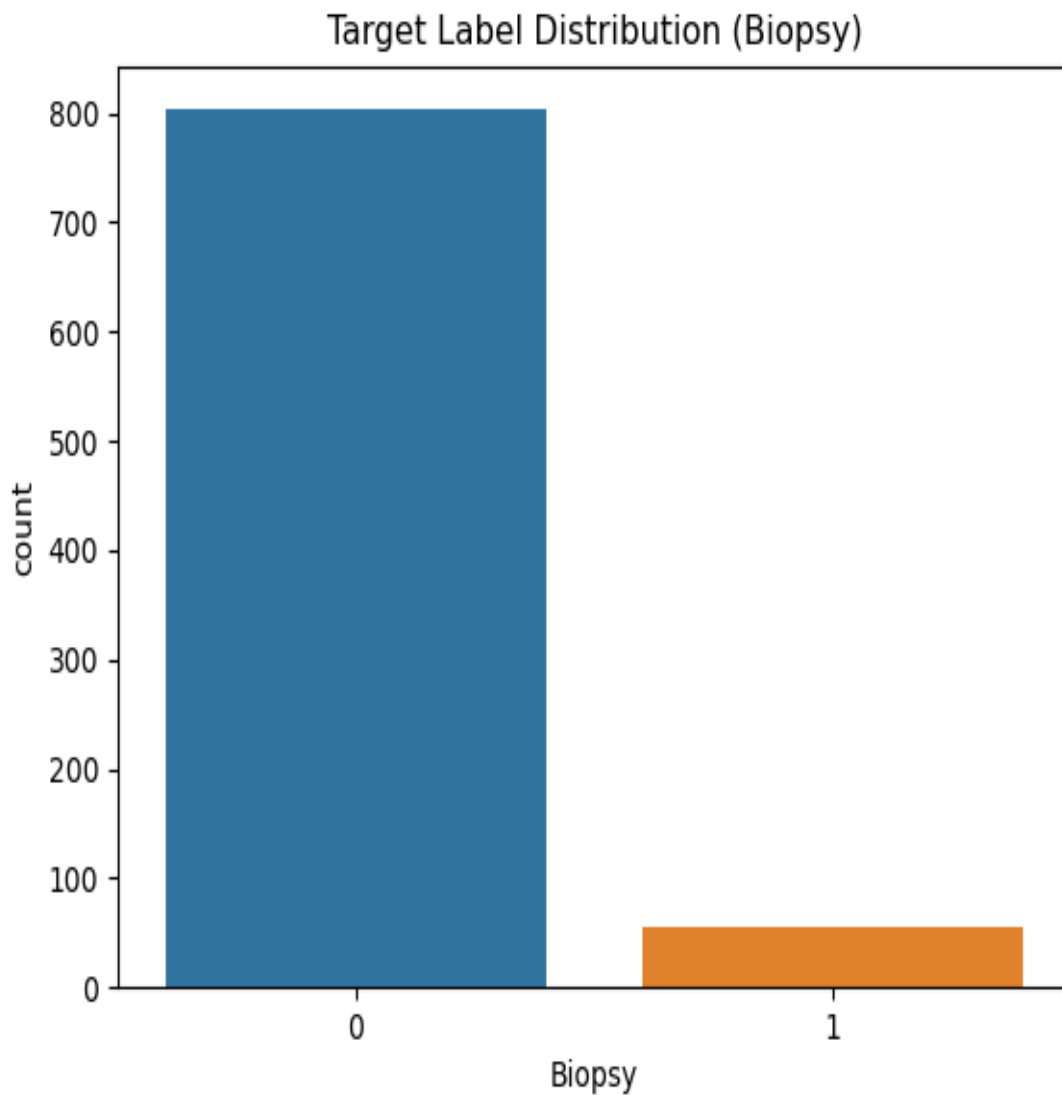


Target Variable Distribution

The Biopsy variable shows a class imbalance:

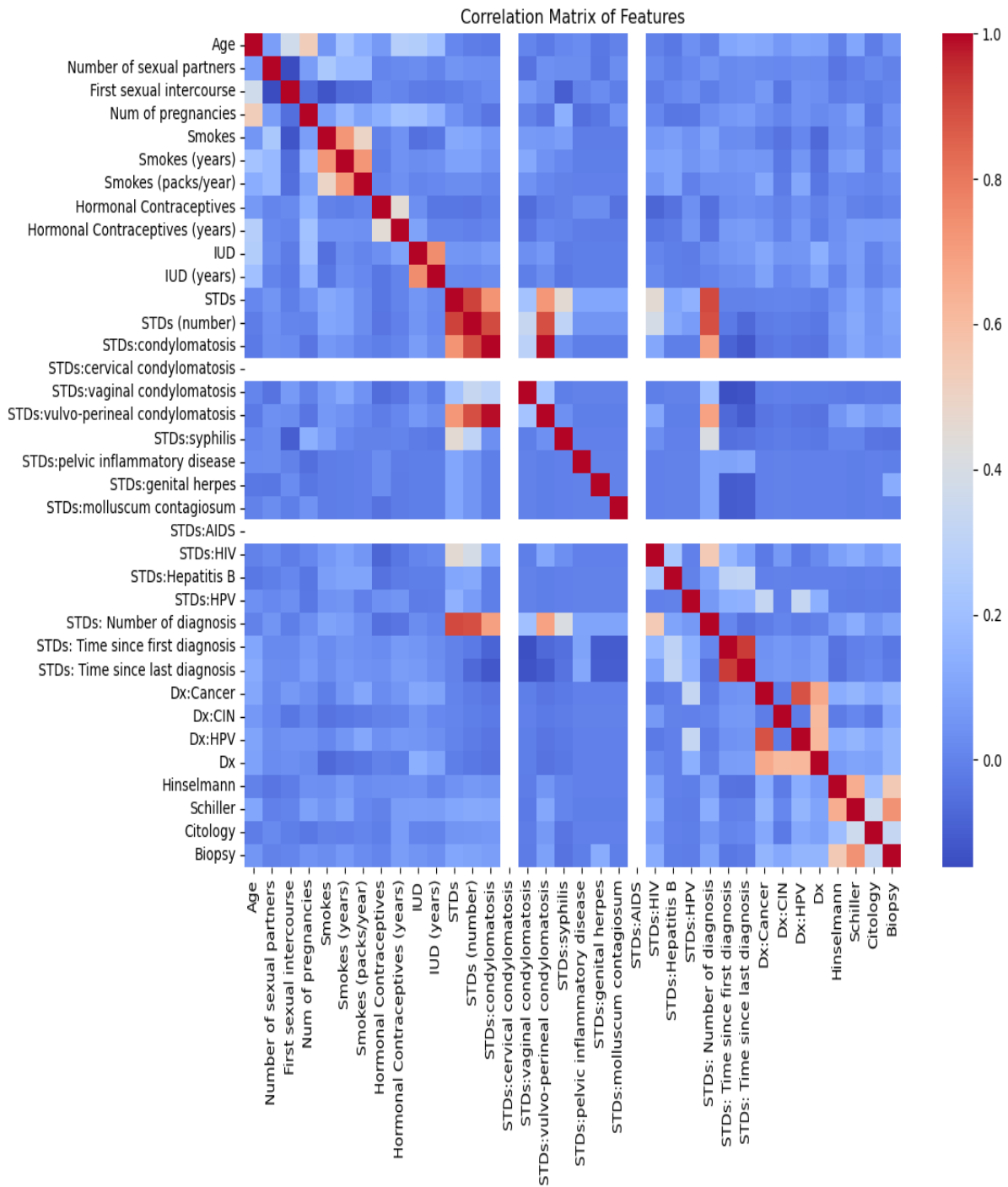
- 93.5% of the samples are negative (0)

- 6.5% are positive (1)



Correlation Matrix Insights

- Smoking duration is highly correlated with the number of packs per year.
- Hormonal contraceptive use is moderately correlated with years of usage.
- Most diagnosis-related variables show low correlation with the target variable.

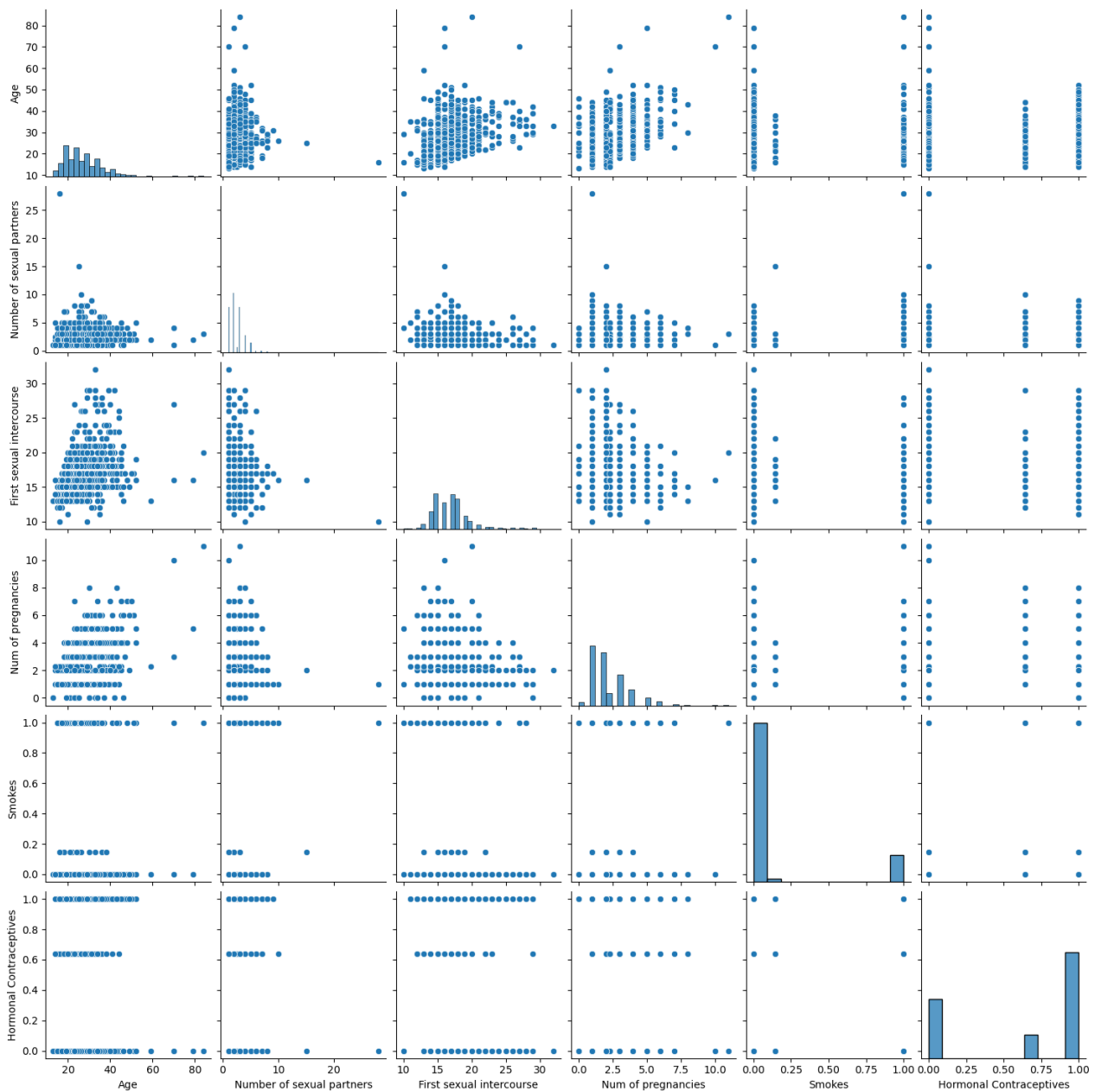


Pairplot Analysis

A pairplot of selected features (e.g., Age, Number of sexual partners, First sexual intercourse, Smokes, Biopsy) reveals patterns:

- Positive cases cluster around older ages and higher number of sexual partners.

- Smokers appear more frequently among positive biopsy outcomes.

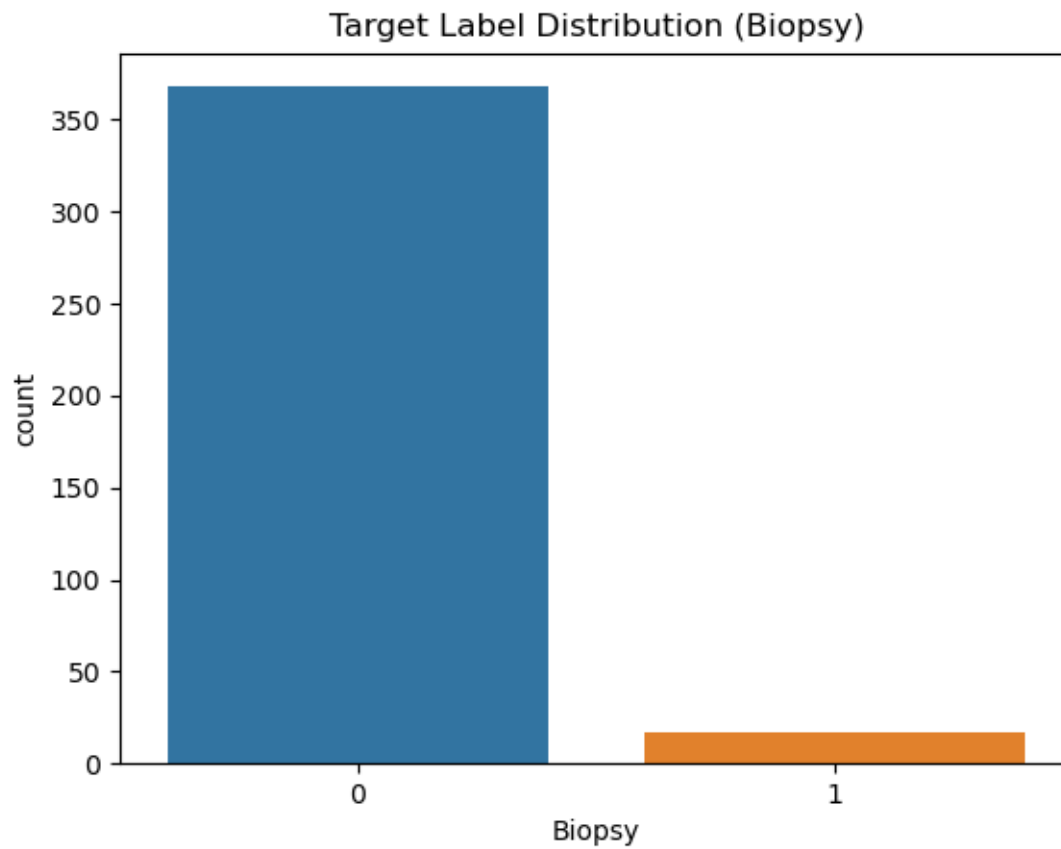


Cleaned Dataset Overview

After outlier removal:

- Final dataset has **385 rows**
- All 36 features are retained

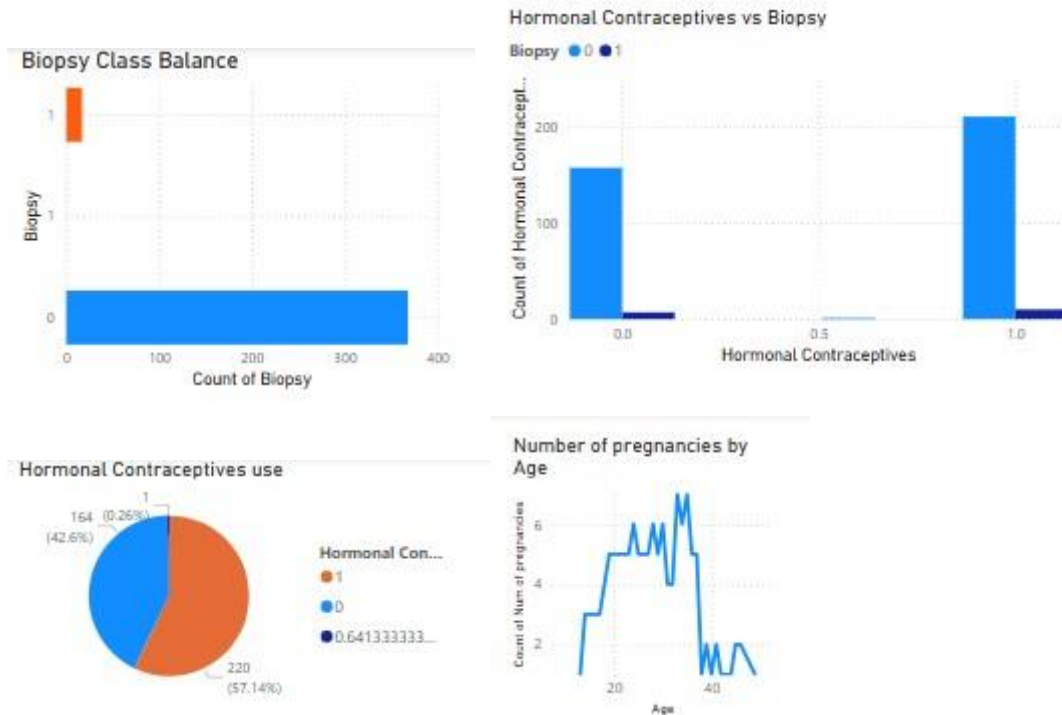
- Dataset is cleaner and better suited for predictive modeling



5. Data Visualization Using Power BI

Power BI was used to generate interactive dashboards including:

- Bar chart of Biopsy outcome distribution
- Slicers for contraceptive use and age group analysis



6. Conclusion

The descriptive analysis of the cervical cancer risk factors dataset reveals meaningful patterns that can support early detection strategies. Though the dataset initially had class imbalance and noise, the revised preprocessing steps improved its quality. Insights derived can guide further predictive analytics and help build decision support tools for rural healthcare systems.

7. References

1. UCI Machine Learning Repository – Cervical Cancer (Risk Factors) Dataset: <https://archive.ics.uci.edu/ml/datasets/Cervical+Cancer+%28Risk+Factors%29>
2. Power BI Documentation – <https://learn.microsoft.com/en-us/power-bi/>